Classical Field Theory — Exercise Sheet 12

Completing transverse reduction

Solutions

Use the unitary Fourier transform of Lecture 4. The real canonical form is

$$\Omega((a,p),(b,q)) = \int (a\cdot q - b\cdot p).$$

The projector is $P_{\mathrm{T}}(k) = I - kk^{\mathsf{T}}/|k|^2$ for $k \neq 0$. All claims about a fixed annulus below concern the data and smearings specified in Lecture 12, not pointwise Fourier observables.

Exercise 12.1 (A finite Dirac bracket and an oblique mode). For a fixed $k \in \mathbb{R}^3 \setminus \{0\}$, consider the finite canonical model $(a,p) \in \mathbb{R}^3 \times \mathbb{R}^3$ with $\{a_i, p_j\} = \delta_{ij}$. Set $c_1 = k\cdot p$ and $c_2 = k\cdot a$. Given an invertible constraint matrix $C_{ab} = \{c_a, c_b\}$, its *Dirac bracket* is

$$\{F,G\}_D = \{F,G\} - \{F,c_a\}(C^{-1})_{ab}\{c_b,G\}.$$

This formula is supplied for this finite system.

(a) Compute C, its inverse, and every coordinate Dirac bracket. Check that each constraint has zero Dirac bracket with every coordinate.

(b) At $k = (1,2,2)$, display the projector and project $(3,-1,2)$. Recover the annulus smeared bracket of Lecture 12 from this matrix symbol, using real vector smearings f, g.

(c) At $k = 0$, which step fails? Explain why periodic real gauge parameters on $\mathbb{T}^3$ leave three constant canonical pairs, and give their form.

Solution. For the first part, the two scalar constraints give

$$C = \begin{pmatrix} 0 & -|k|^2 \\ |k|^2 & 0 \end{pmatrix}, \qquad C^{-1} = \begin{pmatrix} 0 & |k|^{-2} \\ -|k|^{-2} & 0 \end{pmatrix}.$$

Only $\{a_i, c_1\} = k_i$ and $\{c_2, p_j\} = k_j$ contribute to the mixed coordinate correction. Thus

$$\{a_i,p_j\}_D = \delta_{ij} - \frac{k_i k_j}{|k|^2}, \qquad \{a_i,a_j\}_D = \{p_i,p_j\}_D = 0.$$

Contracting either index with k proves the claimed zero brackets with c_1 and c_2. In this finite model the Gauss direction and a spatial gauge-fixing condition have removed the longitudinal pair.

For the second part, $|k|^2 = 9$ gives

$$P_{\mathrm{T}}(k) = \frac{1}{9}\begin{pmatrix} 8 & -2 & -2 \\ -2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix}, \qquad P_{\mathrm{T}}(k)\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 22 \\ -19 \\ 8 \end{pmatrix}.$$

The dot product of this result with $(1,2,2)$ is zero. For real annulus smears, the gradients of $A_T(f)$ and $\Pi_T(g)$ are $(P_{\mathrm{T}}f, 0)$ and $(0, P_{\mathrm{T}}g)$. Plancherel and orthogonality therefore give

$$\{A_T(f), \Pi_T(g)\}_{\mathrm{red}} = \int P_{\mathrm{T}}f \cdot P_{\mathrm{T}}g = \int f \cdot P_{\mathrm{T}}g.$$

This is the tested matrix symbol. The finite model does not turn a Fourier point value into an admissible observable, or establish an unrestricted full-frequency delta kernel from one annulus.

For the last part, both constraints become identically zero at $k = 0$. The matrix cannot be inverted there. A periodic real χ has $\int_{\mathbb{T}^3} \nabla\chi = 0$, so it cannot alter a constant potential. Gauss imposes no condition on constant momentum. Integrating the canonical form for constant pairs gives

$$\Omega_0 = (2\pi)^3 \sum_{i=1}^{3} \delta a_i \wedge \delta p_i.$$

All three harmonic pairs remain.

Exercise 12.2 (A union of annuli, rather than one annulus). For $n \geq 2$ let $\mathcal{U}_n = \{n^{-1} < |k| < n\}$ and let $\mathcal{Y}$ be the union of the real Schwartz classes with smooth compact Fourier support in $\mathcal{U}_n$.

(a) Prove that $\mathcal{Y}$ is dense in the real spaces $L^2(\mathbb{R}^3;\mathbb{R})$ and $H^1(\mathbb{R}^3;\mathbb{R})$, using the appropriate norm in each case. You may use the usual L^2 density of compact smooth functions, proved by truncation and smoothing.

(b) Prove the corresponding density for transverse vector fields, with approximants $P_{\mathrm{T}} v_n$. Keep the real-field Fourier condition.

(c) Show that the analogous class in one fixed annulus is not dense in either whole space. Explain why a bounded extension from a dense union must not be justified by density of just one annulus.

Solution. For the first part, write $w_s(k) = (1 + |k|^2)^s$ with $s = 0$ or $s = 1$. Choose real even smooth cutoffs η_n supported in $\mathcal{U}_n$, between zero and one, and tending to one at every $k \neq 0$. Dominated convergence gives

$$\int w_s |\eta_n \widehat{f} - \widehat{f}|^2 \mathrm{d}^3k \longrightarrow 0.$$

At fixed n, approximate $\eta_n \widehat{f}$ by smooth compact profiles inside $\mathcal{U}_n$. This can be done by first cutting away any remaining boundary strip and then convolving with a sufficiently small mollifier. On that bounded set the weighted and ordinary L^2 norms are equivalent. Choose the approximation error at most $1/n$ in the weighted norm. Replace its profile b_n by

$$\frac{1}{2}\big(b_n(k) + \overline{b_n(-k)}\big).$$

This preserves its annulus support, gives Fourier reality, and does not increase its error from the real target. Inverse transformation proves the stated density in the corresponding spatial norm.

Here H^0 means L^2. For the second part, apply that construction componentwise to a transverse v. The multiplier is real and even, so $P_{\mathrm{T}} v_n$ remains real with the same smooth annulus support. Its contraction estimate with either weight is

$$\|P_{\mathrm{T}}(v_n - v)\|_{H^s}^2 = \int w_s |P_{\mathrm{T}}(k)(\widehat{v}_n - \widehat{v})|^2 \mathrm{d}^3k \leq \|v_n - v\|_{H^s}^2.$$

Since $P_{\mathrm{T}} v = v$, the projected approximants converge to v.

For the final part, choose a nonzero real even smooth Fourier profile supported in a shell disjoint from the fixed annulus. Normalize its inverse transform to have norm one in the

space being tested. Every fixed-annulus profile is orthogonal to it in the weighted Fourier inner product. Consequently its distance from every such profile is at least one. One annulus is not dense. The direct global multiplier in Lecture 12, or extension from the dense union just proved, avoids this error.

Exercise 12.3 (A gauge parameter outside L^2). Choose a real radial $\eta \in C_c^\infty(\mathbb{R}^3)$ equal to one on $|k| \le 1$. Define a tempered scalar distribution and a vector by their Fourier profiles

$$\widehat{\chi}_0(k) = \frac{\eta(k)}{|k|^2}, \qquad \widehat{A}(k) = ik\,\widehat{\chi}_0(k) \quad (k \neq 0).$$

A locally integrable function of at most polynomial growth defines a tempered distribution. You may also use that a distribution with all first derivatives zero is constant.

(a) Show that χ_0 is well-defined and real, that $A \in H^1$, and that $\chi_0 \notin L^2$. Can any L^2 scalar have gradient A?

(b) For a general real $A \in H^1(\mathbb{R}^3;\mathbb{R}^3)$ compare the two profiles $\widehat{\chi}_L = -i(k\cdot\widehat{A})/|k|^2$ and $-\widehat{\chi}_L$. Which one has gradient $(I-P_{\mathrm{T}})A$, and which gauges A to $P_{\mathrm{T}}A$? Justify that χ_L is tempered even near zero frequency.

Solution. For the first part, r^{-2} is locally integrable in three dimensions because $\int_0^1 r^2 r^{-2}\mathrm{d}r < \infty$. Its compact support makes the resulting distribution tempered. The profile is real and even, so the distribution is real. The vector profile is imaginary and odd, which is the Fourier reality condition for A. The squared parameter profile has radial integral $\int_0^1 r^{-2}\mathrm{d}r = \infty$. Plancherel therefore rules out $\chi_0 \in L^2$. For its gradient, the weighted radial integral near zero is instead

$$\int_0^1 (1+r^2) r^2 r^{-2}\mathrm{d}r < \infty.$$

There is no large-frequency issue because η is compactly supported. Thus $A \in H^1$. If an L^2 parameter had this gradient, it would differ from χ_0 by a constant distribution. In Fourier space a nonzero constant adds a point delta at zero, which cannot be an L^2 function; a zero constant leaves the non-L^2 profile η/r^2. Neither gives an L^2 parameter.

For the second part, direct multiplication gives

$$ik\widehat{\chi}_L = \frac{k(k\cdot\widehat{A})}{|k|^2} = (I-P_{\mathrm{T}})\widehat{A}, \qquad \widehat{A} + ik(-\widehat{\chi}_L) = P_{\mathrm{T}}\widehat{A}.$$

The two signs serve different purposes. Fourier reality follows by replacing k by $-k$ and conjugating $\widehat{A}$. Near zero the bound $|\widehat{\chi}_L| \le |\widehat{A}|/|k|$ gives

$$\int_{|k|<1} |\widehat{\chi}_L|\mathrm{d}^3k \le \|\widehat{A}\|_2 \left(\int_{|k|<1} |k|^{-2}\mathrm{d}^3k\right)^{1/2} < \infty.$$

At infinity its pairing with any Schwartz test is finite by Cauchy–Schwarz. This defines a tempered distribution with gradient in H^1. The completion therefore uses distributional parameters modulo constants, not just Schwartz or H^1 scalar parameters.

Exercise 12.4 (Weak symplectic does not mean every gradient exists). On the real space $\mathcal{E}_T = H^1_{\mathrm{trans}} \times L^2_{\mathrm{trans}}$, use the canonical form from Lecture 12. Let $\theta(r)$ be smooth, zero for $r \le 1$ and one for $r \ge 2$, and define the real transverse L^2 vector h by

$$\widehat{h}(k) = \theta(|k|)|k|^{-2} P_{\mathrm{T}}(k) e_1 .$$

(a) Prove that $h \in L^2$ but $h \notin H^1$. Show that $F(a,p) = \int h \cdot p \, \mathrm{d}^3x$ is continuous on $\mathcal{E}_T$.

(b) Show that no $X \in \mathcal{E}_T$ satisfies $\iota_X \Omega = DF$. Why does this not contradict the nondegeneracy proved in Lecture 12?

(c) Does finiteness of $\int (|\operatorname{curl} A|^2 + |\Pi|^2)$ imply membership in this inhomogeneous phase space? Test a nonzero constant potential with zero momentum.

Solution. For the first part, $P_{\mathrm{T}}(n)e_1$ has squared norm $1 - n_1^2$. Its angular integral is $8\pi/3 > 0$. For $r \ge 2$, the squared L^2 and H^1 norms therefore contain the radial factors

$$\int_2^\infty r^{-2} \mathrm{d}r < \infty, \qquad \int_2^\infty (1 + r^2) r^{-2} \mathrm{d}r = \infty .$$

The transition region is harmless. The profile is real and even, so h is real, and $P_{\mathrm{T}} h = h$ makes it transverse. Cauchy–Schwarz gives

$$|F(a,p)| \le \|h\|_2 \|p\|_2 ,$$

which proves continuity on the stated phase space.

For the second part, suppose $X = (X_A, X_\Pi)$ represents DF. Testing its contraction against $(0, q)$ gives

$$\int X_A \cdot q = \int h \cdot q \qquad (q \in L^2_{\mathrm{trans}}).$$

Choose $q = X_A - h$, an allowed transverse L^2 vector. Then $\|X_A - h\|_2^2 = 0$, so $X_A = h$, contradicting $X_A \in H^1$. Nondegeneracy only states that a vector pairing to zero with every tangent is zero. It does not assert that every continuous covector has a representing vector. This is the distinction expressed by *weakly symplectic*.

For the final part, take $A = e_1$ and $\Pi = 0$ on all of $\mathbb{R}^3$. Its curl and physical energy vanish, but $A \notin L^2$. It therefore lies outside $H^1_{\mathrm{trans}} \times L^2_{\mathrm{trans}}$. The inhomogeneous completion is a specified phase model; finite physical energy alone does not justify its canonical integral or its claimed domain.